

Maulana Bisyr Azhari

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Education

Master-PhD Integrated in Electrical Engineering

Korea Advanced Institute of Science and Technology (KAIST)

- **Advisor** : Prof. David Hyunchul Shim.
- **Topics** : Robotics, Autonomous Drone, Localization, Computer Vision.

Current GPA - 3.54/4.3

08/2021 - Current

Bachelor of Engineering in Electrical Engineering

University of Indonesia

- **Advisor** : Prof. Benyamin Kusumoputro.
- **Thesis** : Missile Dynamical System Identification using Artificial Neural Networks.
- **Courses** : Control Engineering, Deep Learning, Robotics.

GPA - 3.53/4.0

08/2016 - 07/2020

Experiences

Unmanned System Research Group (USRG) - KAIST EE

Graduate Student Researcher

- Develop real-world drones for various applications: Search-rescue, reconnaissance, anti-drone, maritime security, and autonomous drone racing.
- Robotics stacks: Autonomous navigation, localization, detection, autonomous exploration, control.

Daejeon, South Korea

Aug 2021 - Present

PT Indo Xetia Internasional (XETIA)

Backend/Machine Learning Engineer

- Developed microservices for user management, real-time chat, notification, product, and AI models. Stacks: Django, REST Framework, WebSocket, FastAPI, Redis, Django channels, Celery, Flask, TensorFlow-serving, Docker, Dash, Paddle-Serving
- Designed and developed machine learning models for various applications: Image-based categorization, clustering, etc

Remote, Indonesia

Sep 2020 - Aug 2021

Computational Intelligence and Intelligent Systems Laboratory – University of Indonesia

Research Assistant

- Conducted research in control system algorithm and computer vision. Assisted undergraduate students in their research.

Depok, Indonesia

Aug 2020 - Aug 2021

Indonesia Medical Education and Research Institute (IMERI FKUI)

Medical Technology Research and Development Intern

- Designed and implemented a PID controller to Negative Pressure Wound Therapy (NPWT) Device.
- Designed and fabricated PCB for the NPWT.

Jakarta, Indonesia

Oct 2019 - Nov 2019

PT Industri Kereta Api (INKA) - State-owned Rolling Stock Manufacturer

Technology Research and Development Intern

- Designed and simulated an Engine-Generator Control Unit of an Electric-Diesel Train.
- Designed the electrical components of the system such as rectifier, low pass filter, and buck converter.

Madiun, Indonesia

Jul 2019 - Aug 2019

Publications

JOURNALS

1. **M.B. Azhari**, and D.H. Shim. "DINO-VO: A Feature-based Visual Odometry Leveraging a Visual Foundation Model," IEEE Robotics and Automation Letters, July 2025. [\[paper\]](#)
2. B. Kim, **M.B. Azhari**, J. Park, and D.H. Shim. "An autonomous UAV system based on adaptive LiDAR Inertial Odometry for practical exploration in complex environments," Journal of Field Robotics, 2024. [\[paper\]](#) [\[video1\]](#) [\[video2\]](#)
3. J. Park, B. Kim, S. Lee, **M.B. Azhari** and D.H. Shim, "Curvature-based 3D Path Planning Algorithm for Quadcopter," The Journal of Korea Robotics Society, vol.18, no.3, pp. 316-322. 2023. [\[paper\]](#)

CONFERENCES

1. G. Kang, O. Guenes, S. Lee, **M.B. Azhari**, D.H. Shim. "SPIBOT: A Drone-Tethered Mobile Gripper for Robust Aerial Object Retrieval in Dynamic Environments." IEEE International Conference on Robotics and Automation (ICRA) 2025. [\[paper\]](#) [\[video\]](#)
2. M.F. Fadlian, **M.B. Azhari**, B. Kusumoputro. "Data acquisition of X-plane's aircraft through matlab for neural network based identification system." AIP Conf. Proc. 23 September 2021; 2376 (1): 060002. [\[paper\]](#)

PREPRINTS/UNDER REVIEW

1. **M.B. Azhari**, D. Han, S. Park, and D.H. Shim. "Robust Tightly-Coupled Filter-based Monocular Visual-Inertial State Estimation and Graph-based Evaluation for Autonomous Drone Racing," Submitted to IEEE IROS 2026. [\[paper\]](#)
2. **M.B. Azhari**, D. Han, J. You, S. Park, and D.H. Shim. "Drift-Corrected Monocular VIO and Perception-Aware Planning for Autonomous Drone Racing," under review at Journal of Field Robotics. [\[paper\]](#) [\[video\]](#)
3. S. Han, H. Do, S. Lee, K. Park, **M.B. Azhari**, H. Cho, J. Choi, O. Guenes, D. Han, G. Kang, D. Kim, K. Kim, D. Lee, J. Park, H. Son, J. Kim, D.H. Shim, S. Cho. "A Collaborative USV-UAV System for Maritime Search and Retrieval in GNSS-denied Environment: Team KAIST's Contribution to MBZIRC 2023," under review. [\[video\]](#)
4. S. Lee, **M.B. Azhari**, G. Kang, O. Guenes, D. Han, D.H. Shim. "A Collaborative Team of UAV-Hexapod for an Autonomous Retrieval System in GNSS-Denied Maritime Environments," under review. [\[paper\]](#) [\[video\]](#)

Projects

COMPETITIONS

A2RL Autonomous Drone Championship (2025-2026)

Abu Dhabi, United Arab Emirates

Held by ASPIRE - Abu Dhabi Autonomous Racing League, United Arab Emirates

Sept 2024 - Present

- Goal: Develop algorithms for high-speed drone racing that rely exclusively on a single monocular RGB camera and an IMU sensor for real-time perception, navigation, and control.
- Stacks: Visual-Inertial-based Localization, Gate Detection, Model Predictive Control, Optimal Trajectory Planning.
- Related links: a2rl.io/autonomous-drone-race, https://youtu.be/x_IDrf6j0qs, youtu.be/omnncgCLqqWA

3rd Mohamed Bin Zayed International Robotics Challenge (MBZIRC)

Abu Dhabi, United Arab Emirates

Held by ASPIRE, United Arab Emirates

March 2023 - February 2024

- Goal: The teams developed autonomous drones (UAVs) and an unmanned surface vehicle (USV) to perform inspection task (detect illegal vessels) and intervention task (retrieve stolen objects from illegal vessels) in a GPS-denied marine environment, addressing issues like piracy and smuggling.
- Stacks: Localization (AprilTag-based, Optical Flow), Mission Planner, Control, Object Detection and Tracking.
- Related links: www.mbzirc.com, youtu.be/Ya1pfWYRMrc, youtu.be/FSxC_kbl7EY

ADD Autonomous Drone for Battlefield Environments

Daejeon, South Korea

Held by Republic of Korea Agency for Defense Development (ADD)

October 2022

- Goal: Develop a fully autonomous drone system for reconnaissance in a battlefield indoor environment. The drone had to perform autonomous localization, obstacle avoidance, exploration along with target detection using onboard computation.
- Stacks: Autonomous Exploration Algorithm, Object Detection-Localization (Euclidean Clustering), LiDAR SLAM.
- Related links: youtu.be/h_PsFVhhAs

5th Army Tiger DroneBot Mission Challenge: Indoor Reconnaissance Drone

Daejeon, South Korea

Held by Republic of Korea Army Training and Doctrine Command (ROKA TRADOC)

Aug 2022

- Goal: Develop a fully autonomous drone system that entered a building via a second-floor window, locate hidden objects in real time, and return to its starting point.
- Stacks: Autonomous Exploration Algorithm, Object Detection, Synthetic Dataset Generation, Real-time Video Transmission (Wire-Guard, LTE Module, Google Cloud, ZMQ), LiDAR SLAM.
- Related links: youtu.be/SXe_FJpxv94

LABORATORY PROJECTS

Underground Autonomous Exploration using Quadrotor UAV

Dec 2022 - Present

Funded by

Indoor Decontamination System using UGV and UAV

April 2024 - Present

Funded by

Anti-Drone System with Jammer

Dec 2022 - Present

Funded by

- Designed and implemented **C++ ROS** packages to autonomously navigate using **Dijkstra's** and to perform robotic manipulation.

Achievements

2026	KAIST Challenge Award , KAIST's 55th Anniversary	Daejeon, South Korea
2026	4th Place of Speed Race , A2RL Drone Championship	Abu Dhabi, United Arab Emirates
2025	3rd Place of AI Grand Challenge , A2RL X DCL Drone Championship	Abu Dhabi, United Arab Emirates
2025	2nd Place of AI Multi-Drone Race , A2RL X DCL Drone Championship	Abu Dhabi, United Arab Emirates
2025	2nd Place of AI Drag Race , A2RL X DCL Drone Championship	Abu Dhabi, United Arab Emirates
2025	KAIST Challenge Award , KAIST's 54th Anniversary	Daejeon, South Korea
2024	2nd Place , 3rd Mohamed Bin Zayed Robotics Challenge (MBZIRC)	Abu Dhabi, United Arab Emirates
2022	Top 2 Teams , ADD Autonomous Drone for Battelfield Environments	Daejeon, South Korea
2022	1st Place , 5th Army Tiger DroneBot Challenge: Indoor Reconnaissance Drone	Daejeon, South Korea
2019	Finalist of EDF Rocket Control System , KOMURINDO-KOMBAT	Garut, Indonesia
2018	Finalist , 31st National Student Scientific Week (PIMNAS)	Yogyakarta, Indonesia
2016	BIDIKMISI Scholarship , Ministry of Research, Technology, and Higher Education	Depok, Indonesia
2015	1st Place of Quick and Smart , ESA FEST at Sultan Ageng Tirtayasa University	Serang, Indonesia

Skills

Programming	Python, C/C++, C#, Matlab
Robotics	ROS1, ROS2, Gazebo, PX4-Autopilot, Localization, Navigation, Control Systems, Embedded Systems, Deep Learning, Perception, Control, Object Tracking
Software	Linux, Tensorflow, Pytorch, Docker, OpenCV, AWS, Google-Cloud, Django, REST API
Certifications	Deep Learning Engineer (Workera, Nov 2020), TensorFlow: Data and Deployment (DeepLearning.AI – Coursera, Oct 2020), Reinforcement Learning (University of Alberta – Coursera, Apr 2020)
Languages	English (Professional), Bahasa Indonesia (Native)